25. The values of ΔH and ΔS for a reaction are respectively 30 kJ mol^{-1} and 100 JK^{-1} mol^{-1}. Then the temperature above which the reaction will become spontaneous is

a) 300 K b) 30 K c) 100 K d) 20^0 C

II. Write brief answer to the following questions:

26. State the first law of thermodynamics.

27. Define Hess's law of constant heat summation.

28. Explain intensive properties with two examples

29. Define the following terms:

a. isothermal process b. adiabatic process

c. isobaric process d. isochoric process

30. What is the usual definition of entropy? What is the unit of entropy?

31. Predict the feasibility of a reaction when

i) both ΔH and ΔS positive ii) both ΔH and ΔS negative

iii) ΔH decreases but ΔS increases

32. Define is Gibb's free energy.

33. Define enthalpy of combustion.

34. Define molar heat capacity. Give its unit.

35. Define the calorific value of food. What is the unit of calorific value?

36. Define enthalpy of neutralization.

37. What is lattice energy?

38. What are state and path functions? Give two examples.

39. Give Kelvin statement of second law of thermodynamics.

40. The equilibrium constant of a reaction is 10, what will be the sign of ΔG? Will this reaction be spontaneous?

41. Enthalpy of neutralization is always a constant when a strong acid is neutralized by a strong base: account for the statement.

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